

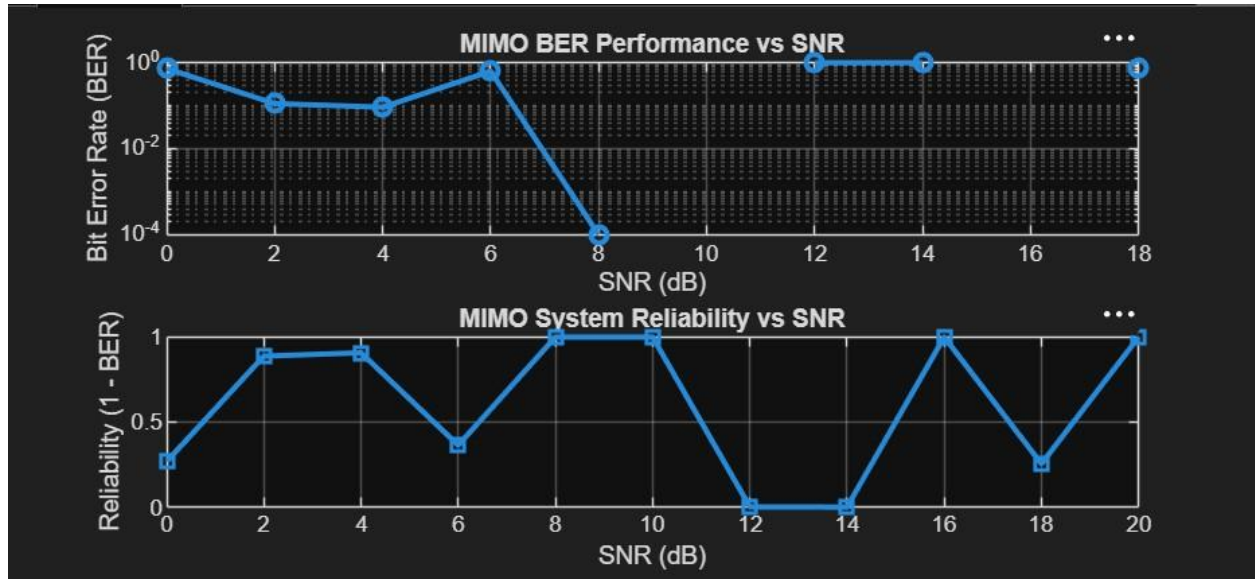
EXP 5

T.Mathivanan
192412205

OBJECTIVE 1

CODE:

```
clc; clear; close all ;
% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
BER =
zeros(1,length(SNR)); for i
= 1:length(SNR) bits =
randi([0 1],10000,1);
symbols = 2*bits-1; %
BPSK
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2); noise =
(randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i)); tx =
H(1,1)*symbols' + noise(1,:); rx = real(tx) > 0;
BER(i) = sum(rx' ~=
bits)/length(bits); end figure
% ----- Graph 1: BER vs SNR -----
subplot(2,1,1) semilogy(SNR_dB,BER, '-
o' , 'LineWidth' ,2) title( 'MIMO BER
Performance vs SNR' ) xlabel( 'SNR (dB)'
) ylabel( 'Bit Error Rate (BER)' ) grid on
% ----- Graph 2: SNR vs Quality Indicator -----
- subplot(2,1,2) plot(SNR_dB,1-BER, '-s' ,
'LineWidth' ,2) title( 'MIMO System Reliability vs
SNR' ) xlabel( 'SNR (dB)' ) ylabel( 'Reliability (1 -
BER)' ) grid on
OUTPUT:
```



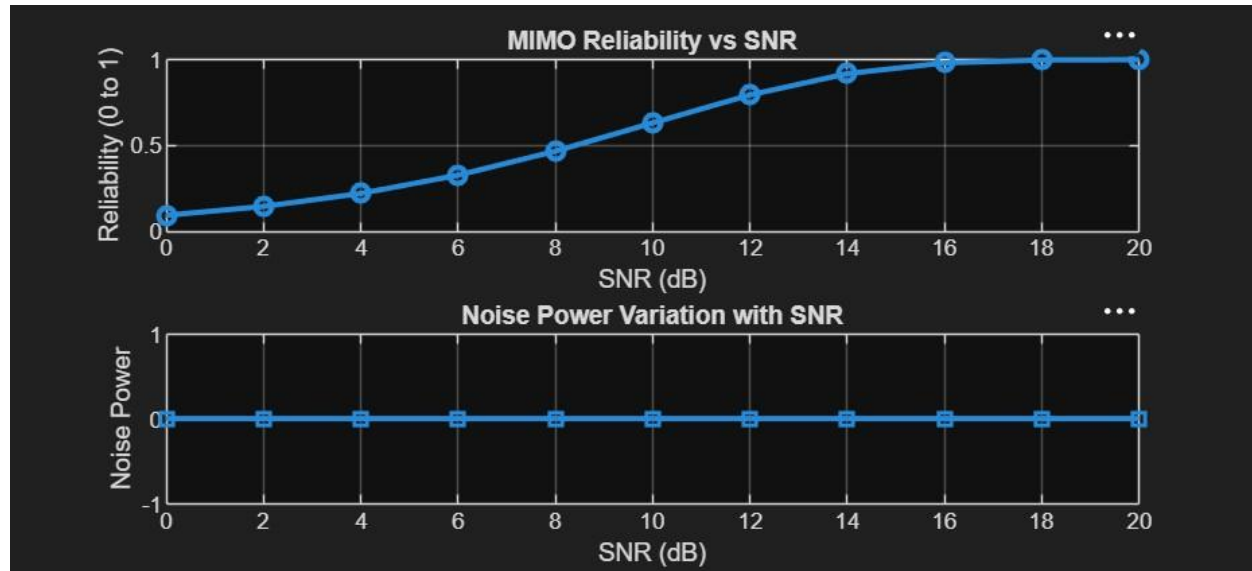
OBJECTIVE 2

CODE:

```
clc; clear; close all ;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER
model) Reliability = zeros(1,length(SNR)); for i =
1:length(SNR)
% simple channel model effect H =
(randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i); % reliability
increases with SNR Reliability(i) = 1 - exp(-
SNR(i)/10); end figure
% ----- Graph 1: SNR vs Reliability -----
- subplot(2,1,1) plot(SNR_dB, Reliability, '-
o', 'LineWidth', 2) title('MIMO Reliability
vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
```

```
% ----- Graph 2: SNR vs Noise Effect ----- subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth',
2) title('Noise Power Variation with SNR') xlabel('SNR (dB)')
ylabel('Noise Power') grid on
```

OUTPUT:



OBJECTIVE 3

CODE:

```
clc; clear; close all ;
% Resource Block parameters
rb_bw = 180e3; % 180 kHz per RB
% Users
users = 1:5;
% Allocated Resource Blocks per user alloc_RB =
[10 12 8 15 5]; % Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM
etc. % Throughput calculation (Mbps) throughput =
(alloc_RB .* rb_bw .* mod_eff) / 1e6;
% Display results disp('User
Throughput (Mbps):')
disp(throughput') figure
% ----- Graph 1: Throughput per User -----
- subplot(2,1,1) plot(users, throughput, '-o',
```

```

'LineWidth',2) title('OFDMA User
Throughput') xlabel('User Index') ylabel(
'Throughput (Mbps)') grid on
% ----- Graph 2: RB vs Throughput -----
- subplot(2,1,2) plot(alloc_RB, throughput,
'-s', 'LineWidth',2) title('Resource Blocks
vs Throughput') xlabel('Allocated
Resource Blocks') ylabel('Throughput
(Mbps)') grid on

```

OUTPUT:

